

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure Equipment Trace 1610 GC Chromatography Thermo Scientific for DGA

Equipment Number: EQ-22 and EQ-23
Serial Number: 822110041 and 722531083



1. PURPOSE

This standard test method procedure is intended to provide users with the guide on the operation and maintenance of Perkin Elmer Clarus 590 Gas Chromatography.

2. SCOPE

This standard operating procedure is employed in the GLTS Analytical Laboratory primarily for determination of dissolved gasses in transformer oil by Thermoscientific trace 1610 GC, a packed injection port, a headspace mounting bracket three capillary columns, a Thermal Conductivity Detector (TCD), methanizer and a Flame Ionization Detector (FID).

3. TERMS AND DEFINITIONS

Gas Chromatography (GC) is a chromatographic technique that separates a mixture of compounds to identify and purify the individual component of the mixture is gas state.

Headspace is a volume of gas phase in contact with a volume of oil in a closed vessel. The vessel is a headspace vial of 20ml nominal capacity.

4. RESPONSIBILITIES

Use, care and maintenance of the Trace 1610 GC is directly assigned to the end user who is usually the Sr. laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. laboratory chemist. Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

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5. SAFETY PRECAUTIONS

- Use a proper electrical outlet of correct voltage and current handling capacity.
- Disconnect from the power supply prior to maintenance and servicing.
- Ensure that the operating switch operates properly.
- Ensure that all cylinder gases line have no leaks.

6. OPERATIONAL DETAILS

- General Instructions:
 - a. Open all gas cylinders. Make sure that gas pressure is at its required value.
 - b. Turn on the instrument system (PC, GC Unit and Headspace) by switch the on/off button.
 - c. Turn on the PC and open Chromeleon software and connect headspace and Trace 1610 GC to software.
 - d. Click Chromeleon Icon in the screen, Chromeleon Workstation Log in Window will appear.
 - e. The Chromeleon Main Menu Screen appears with menu of commands.
 - f. To take control of the unit or to connect the PC Software to GC utilities, click instrument and select headspace the click “connect”, then select on Trace 1610 and click “connect”.
 - g. Once the entire unit is completely connected, different commands can now be performed like setting up instrument, building method, building sequence, reprocess results and calibration of methods.
- Setting Up Sequence for analysis:

Click the Sequence box.

Load a sample sequence with vial position number and method in excel sheet.

Save the edited sequence with unique identification name.

Click on Action and then on Setup.

From Setup instrument dialog box, in sequence select created sequence.

Uncheck Suppress reports/plot and press OK.

Note1: Make sure all samples are prepared as per ASTM D 3612 & IEC60567 and are in auto sampler vial rack ready for injection.

7. MAINTENANCE

- A yearly service preventive maintenance is provided by Thermoscientific Instrument Engineer.
- Gas Supply:

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- a. Check every time before turning on the instrument that all gasses supplies meet the required pressure.
 - b. Check all cylinder gases for no leaks.
 - c. Column Check Up
 - d. Monitor the column performance.
 - e. If possible, condition the column every month and then at its maximum temperature to clean the contaminants inside.
 - f. Run mixture of gases to check the quality of the separation.
- Calibration and Verification
 - a. For calibration curve, three certified Canada STD 50, 100 and 500 ppm (*as per ASTM D 3612 & IEC60567*)
 - b. Verification of calibration curve by running standards prepared in-house (125, 250, 500 ppm).

REFERENCE

- [1] Chromeleon Software (User's Guide)
- [2] ASTM D3612- Standard test method for analysis of gases dissolved in electrical insulating oil by gas chromatography.
- [3] IEC 60567- Oil-filled electrical equipment- sampling of gases and of oil for analysis of free and dissolved gases- guidance.

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